

Full positive answer to Problem 4.9 of arXiv:2505.19143

Vector-valued Bourgain-Morrey block spaces are norm-solid

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Source and Question

The source paper is Tengfei Bai, Pengfei Guo, and Jingshi Xu, *The preduals of Banach space valued Bourgain-Morrey spaces*, arXiv:2505.19143 [1].

Problem 4.9 asks whether the vector-valued block space has the lattice property under the pointwise norm order. In the notation of the source, let X be a Banach space with a predual $*X$, let $*X$ be reflexive or X be separable, and let

$$1 < p < t < r < \infty \quad \text{or} \quad 1 < p \leq t < r = \infty.$$

If $g \in \mathcal{H}_{p'}^{t',r'}(*X)$ and f is a $*X$ -valued function with

$$\|f(x)\|_{*X} \leq \|g(x)\|_{*X} \quad \text{for a.e. } x,$$

must f also belong to $\mathcal{H}_{p'}^{t',r'}(*X)$?

4.4. Lattice property

In [12, Proposition 2.6], Ho obtained the lattice property of the scalar-valued block spaces. Since there are many spaces X satisfying that $*X$ is reflexive or X be separable, there is no unique criterion to define which is bigger when $x, y \in *X$. For example, when $X = \ell^q, q \in (1, \infty)$. First, we can say $y \in \ell^{q'}$ is not less than $x \in \ell^{q'}$ if $\|x\|_{\ell^{q'}} \leq \|y\|_{\ell^{q'}}$. But, we can also define $y \in \ell^{q'}$ is not less than $x \in \ell^{q'}$ if for each $i \in \mathbb{N}$, $|f_i| \leq |g_i|$. The first way can make us compare all elements in $\ell^{q'}$ but the second be more precise. Since we do not know the structure of X . We use the first way to describe the lattice property of block spaces $\mathcal{H}_{p'}^{t',r'}(*X)$.

We do not show the that block spaces $\mathcal{H}_{p'}^{t',r'}(*X)$ have the lattice property. So a natural question is the following:

Problem 4.9. Let X be a Banach space such that $(*X)^* = X$. Let $*X$ be reflexive or X be separable. Let $1 < p < t < r < \infty$ or $1 < p \leq t < r = \infty$. Let $g \in \mathcal{H}_{p'}^{t',r'}(*X)$. If a $*X$ -valued function f satisfying $\|f(x)\|_{*X} \leq \|g(x)\|_{*X}$ a.e., can we obtain f belongs to $\mathcal{H}_{p'}^{t',r'}(*X)$?

Answer

Yes. In fact, the vector-valued block norm is exactly the scalar block norm of the pointwise Banach norm:

$$\|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)} = \|\|F(\cdot)\|_Y\|_{\mathcal{H}_{p'}^{t',r'}}$$

for every Banach space Y and every strongly measurable Y -valued function F for which either side is finite. Consequently the desired lattice property holds for arbitrary Banach-valued block spaces; no coordinate lattice or order structure on Y is needed.

Proof Intuition

The source's block definition measures a vector-valued block only through the scalar function $x \mapsto \|b(x)\|$. Therefore directions in the Banach space should not matter. One direction is obtained by taking norms in any vector block decomposition. The reverse direction lifts a scalar decomposition of $\|F(\cdot)\|$ by multiplying each scalar block by the measurable unit direction $F(x)/\|F(x)\|$. Solidity then follows by cutting the scalar positive decomposition of $\|g(\cdot)\|$ down to $\|f(\cdot)\|$.

Definitions

We use the source's dyadic block notation. A scalar (p', t') -block is a measurable scalar function a , supported in a dyadic cube Q , such that

$$\|a\|_{L^{p'}} \leq |Q|^{1/t-1/p}.$$

For a Banach space Y , a Y -valued (p', t', Y) -block is a strongly measurable Y -valued function b , supported in a dyadic cube Q , such that

$$\| \|b(\cdot)\|_Y \|_{L^{p'}} \leq |Q|^{1/t-1/p}.$$

The block space $\mathcal{H}_{p'}^{t', r'}(Y)$ consists of those Y -valued functions admitting an almost everywhere convergent representation

$$F = \sum_{(j,k) \in \mathbb{Z}^{n+1}} \lambda_{j,k} b_{j,k}, \quad \{\lambda_{j,k}\} \in \ell^{r'},$$

where $b_{j,k}$ is a (p', t', Y) -block supported on $Q_{j,k}$. Its norm is the infimum of $\|\lambda\|_{\ell^{r'}}$ over all such representations. The scalar space $\mathcal{H}_{p'}^{t', r'}$ is the case $Y = \mathbb{R}$.

Scalar Solidity

Lemma 1 (Scalar solidity). *Let $0 \leq u \leq v$ almost everywhere and $v \in \mathcal{H}_{p'}^{t', r'}$. Then*

$$u \in \mathcal{H}_{p'}^{t', r'}, \quad \|u\|_{\mathcal{H}_{p'}^{t', r'}} \leq \|v\|_{\mathcal{H}_{p'}^{t', r'}}.$$

Proof. Fix $\varepsilon > 0$. Choose an admissible scalar block representation

$$v = \sum_i \lambda_i a_i, \quad \|(\lambda_i)\|_{\ell^{r'}} \leq \|v\|_{\mathcal{H}_{p'}^{t', r'}} + \varepsilon.$$

Here the index i abbreviates (j, k) . Put

$$A(x) = \sum_i |\lambda_i| |a_i(x)|.$$

By the same block-summability argument used in the source for admissible block sums, A is finite almost everywhere and belongs to $\mathcal{H}_{p'}^{t',r'}$ with norm at most $\|(\lambda_i)\|_{\ell^{r'}}$. Since $v \geq 0$ and $v = \sum_i \lambda_i a_i$, the triangle inequality gives the pointwise majorization $0 \leq u \leq v = |v| \leq A$.

Define

$$c_i(x) = \begin{cases} \frac{u(x)}{A(x)} |a_i(x)|, & A(x) > 0, \\ 0, & A(x) = 0. \end{cases}$$

Then $0 \leq c_i \leq |a_i|$, so each c_i is a scalar block supported on the same cube as a_i . Also,

$$u(x) = \sum_i |\lambda_i| c_i(x)$$

for almost every x . Hence u has an admissible block representation with coefficient sequence $(|\lambda_i|)$, and

$$\|u\|_{\mathcal{H}_{p'}^{t',r'}} \leq \|(\lambda_i)\|_{\ell^{r'}} \leq \|v\|_{\mathcal{H}_{p'}^{t',r'}} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Norm Reduction

Lemma 2 (Pointwise norm reduction). *Let Y be a Banach space and let F be a strongly measurable Y -valued function. Then*

$$F \in \mathcal{H}_{p'}^{t',r'}(Y) \iff \|F(\cdot)\|_Y \in \mathcal{H}_{p'}^{t',r'},$$

and in this case

$$\|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)} = \|\|F(\cdot)\|_Y\|_{\mathcal{H}_{p'}^{t',r'}}.$$

Proof. First suppose $F \in \mathcal{H}_{p'}^{t',r'}(Y)$. Fix $\varepsilon > 0$ and choose

$$F = \sum_i \lambda_i b_i, \quad \|(\lambda_i)\|_{\ell^{r'}} \leq \|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)} + \varepsilon,$$

with Y -valued blocks b_i . Set

$$A(x) = \sum_i |\lambda_i| \|b_i(x)\|_Y.$$

The scalar functions $\|b_i(\cdot)\|_Y$ are scalar blocks, so $A \in \mathcal{H}_{p'}^{t',r'}$ with

$$\|A\|_{\mathcal{H}_{p'}^{t',r'}} \leq \|(\lambda_i)\|_{\ell^{r'}}.$$

Moreover $\|F(x)\|_Y \leq A(x)$ almost everywhere. By scalar solidity,

$$\|\|F(\cdot)\|_Y\|_{\mathcal{H}_{p'}^{t',r'}} \leq \|(\lambda_i)\|_{\ell^{r'}} \leq \|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives

$$\|\|F(\cdot)\|_Y\|_{\mathcal{H}_{p'}^{t',r'}} \leq \|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)}.$$

Conversely assume $\phi = \|F(\cdot)\|_Y \in \mathcal{H}_{p'}^{t',r'}$. Define the strongly measurable direction

$$u(x) = \begin{cases} F(x)/\phi(x), & \phi(x) > 0, \\ 0, & \phi(x) = 0. \end{cases}$$

Then $\|u(x)\|_Y \leq 1$ and $F(x) = \phi(x)u(x)$ almost everywhere. Choose an admissible scalar representation

$$\phi = \sum_i \mu_i a_i, \quad \|(\mu_i)\|_{\ell^{r'}} \leq \|\phi\|_{\mathcal{H}_{p'}^{t',r'}} + \varepsilon.$$

For each i , put $B_i(x) = a_i(x)u(x)$. Since

$$\|B_i(x)\|_Y \leq |a_i(x)|,$$

each B_i is a Y -valued block supported on the same cube as a_i . The scalar convergence of $\sum_i \mu_i a_i$ to ϕ implies convergence of $\sum_i \mu_i B_i$ to $\phi u = F$ in Y almost everywhere. Therefore

$$\|F\|_{\mathcal{H}_{p'}^{t',r'}(Y)} \leq \|(\mu_i)\|_{\ell^{r'}} \leq \|\phi\|_{\mathcal{H}_{p'}^{t',r'}} + \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives the reverse inequality and hence the norm identity. $\square$

Main Theorem

Theorem 3 (Affirmative answer to Problem 4.9). *Let X be a Banach space such that $(^*X)^* = X$. Assume *X is reflexive or X is separable, and let*

$$1 < p < t < r < \infty \quad \text{or} \quad 1 < p \leq t < r = \infty.$$

*If $g \in \mathcal{H}_{p'}^{t',r'}(^*X)$ and f is a *X -valued measurable function satisfying*

$$\|f(x)\|_{*X} \leq \|g(x)\|_{*X} \quad \text{for almost every } x \in \mathbb{R}^n,$$

*then $f \in \mathcal{H}_{p'}^{t',r'}(^*X)$. More precisely,*

$$\|f\|_{\mathcal{H}_{p'}^{t',r'}(^*X)} \leq \|g\|_{\mathcal{H}_{p'}^{t',r'}(^*X)}.$$

Proof. Apply the pointwise norm reduction lemma to g . It gives

$$\|g(\cdot)\|_{*X} \in \mathcal{H}_{p'}^{t',r'} \quad \text{and} \quad \|\|g(\cdot)\|_{*X}\|_{\mathcal{H}_{p'}^{t',r'}} = \|g\|_{\mathcal{H}_{p'}^{t',r'}(^*X)}.$$

Since

$$0 \leq \|f(\cdot)\|_{*X} \leq \|g(\cdot)\|_{*X}$$

almost everywhere, scalar solidity gives

$$\|f(\cdot)\|_{*X} \in \mathcal{H}_{p'}^{t',r'}, \quad \|\|f(\cdot)\|_{*X}\|_{\mathcal{H}_{p'}^{t',r'}} \leq \|\|g(\cdot)\|_{*X}\|_{\mathcal{H}_{p'}^{t',r'}}.$$

Applying the pointwise norm reduction lemma again, now to f , yields

$$f \in \mathcal{H}_{p'}^{t',r'}(^*X)$$

and the asserted norm inequality. $\square$

Verification Notes and Limitations

The proof uses only the block definition from the source paper and does not use an order or coordinate lattice on $*X$. Thus it proves the source problem in the stated range and in fact applies to arbitrary Banach targets Y , whenever the corresponding block space is defined with the same pointwise-norm block condition.

No computational verification is needed. The main technical points for human review are:

- the scalar cut-down construction in the scalar solidity lemma;
- measurability of the direction $F/\|F\|$, which follows from strong measurability of F ;
- preservation of block bounds under $a_i \mapsto a_i u$.

Novelty Check

Before promotion, the run indexes were searched for arXiv:2505.19143, the paper title, Bourgain-Morrey block spaces, and lattice-property keywords. No prior packet or attempt was found. The local parsed arXiv corpus was searched for exact phrases from Problem 4.9, including vector-valued Bourgain-Morrey block-space lattice-property phrases; only the source paper surfaced as a direct hit. Web searches on 2026-06-28 for exact phrases including “Problem 4.9 Bourgain-Morrey lattice property” and “Banach space valued Bourgain-Morrey lattice property” returned the source arXiv record and no separate answer in this bounded pass.

References

- [1] T. Bai, P. Guo, and J. Xu, *The preduals of Banach space valued Bourgain-Morrey spaces*, arXiv:2505.19143, 2025.